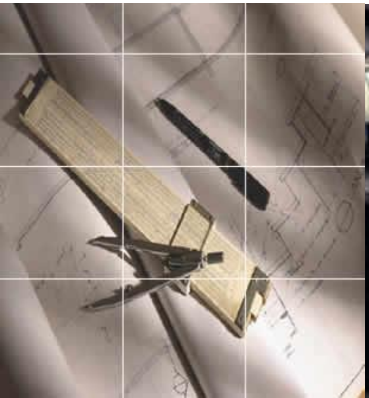
ME-109 Engineering Drawing

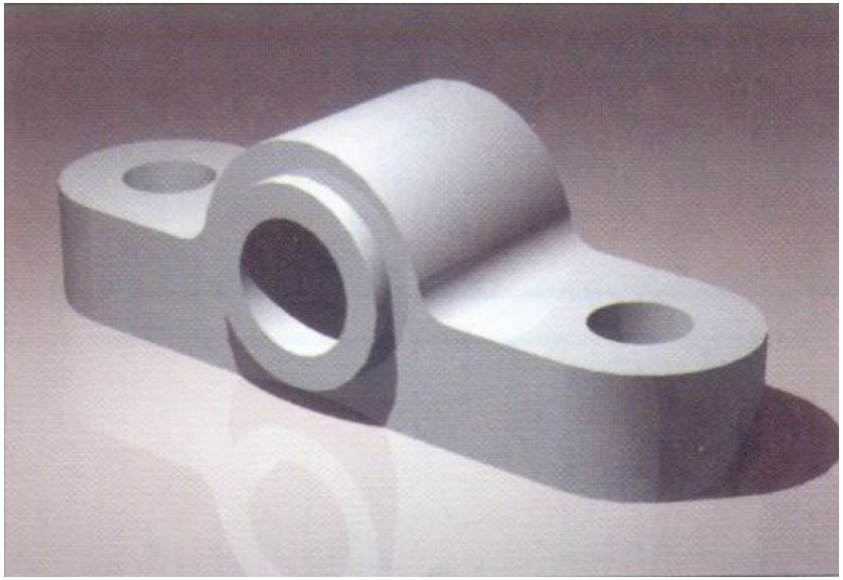
An Introduction




Effectiveness of Graphics Language

1. Try to write a description of this object.

2. Test your written description by having someone attempt to make a sketch from your description.



You can easily understand that …

The word languages are inadequate for describing the ***size***, ***shape*** and ***features*** completely as well as concisely.

Composition of Graphic Language

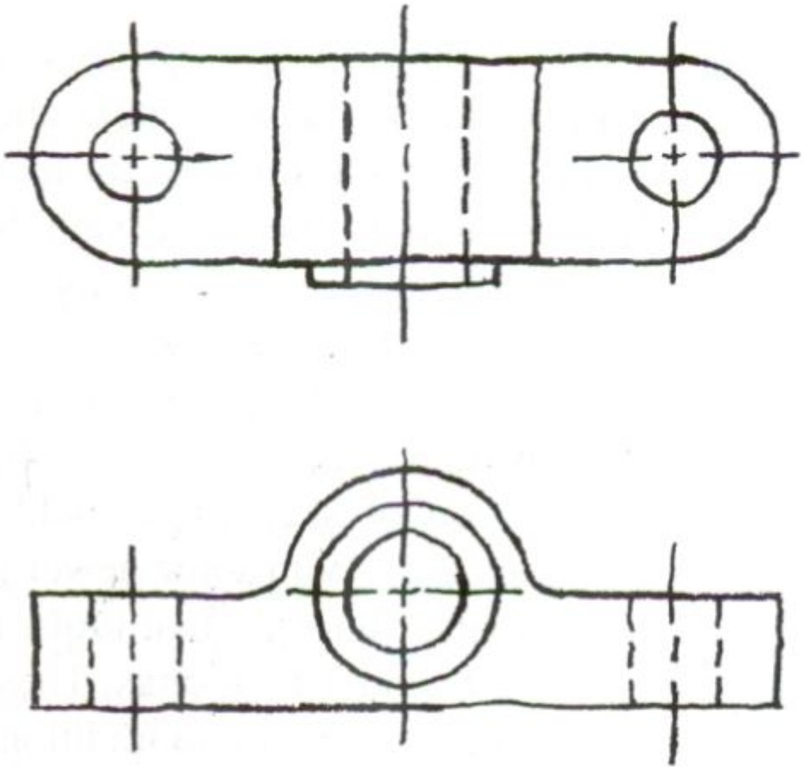
Graphic language in **"engineering application"** use ***lines*** to represent the ***surfaces***, ***edges*** and ***contours*** of objects.

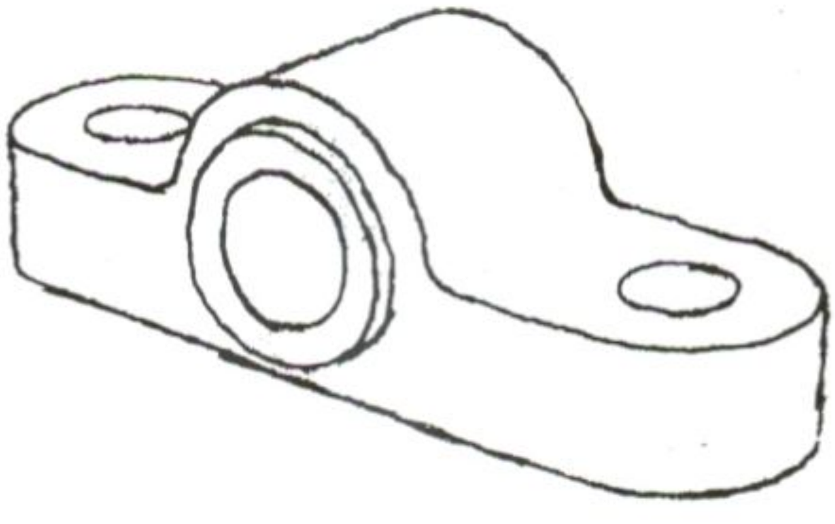
- The language is known as "***drawing***" or "***drafting***" .
- A drawing can be done using ***freehand***, ***instruments*** or ***computer*** methods.

Freehand drawing

The lines are sketched without using instruments other than pencils and erasers.

Example

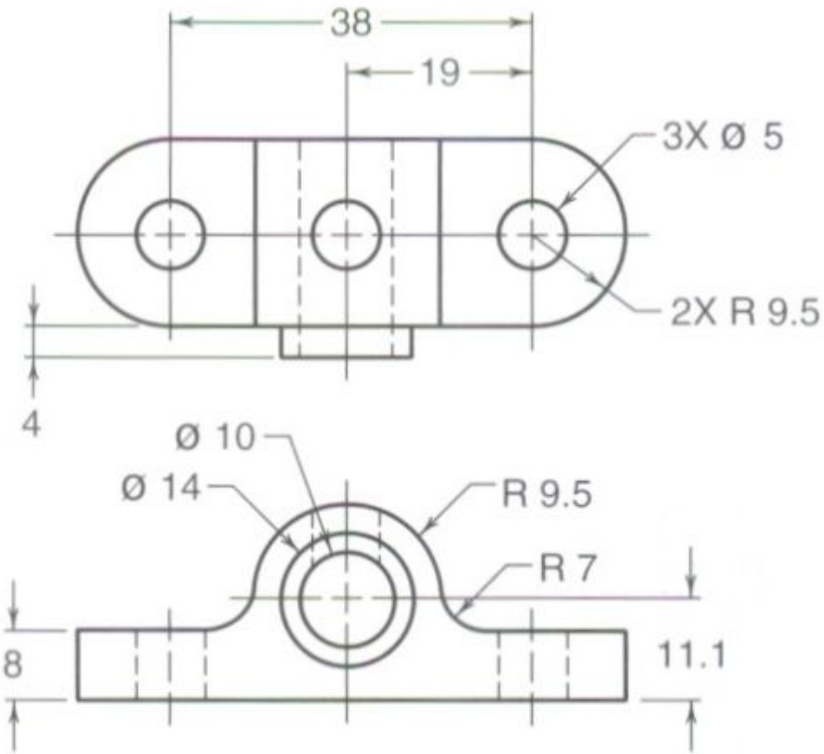




Instrument drawing

Instruments are used to draw straight lines, circles, and curves concisely and accurately. Thus, the drawings are usually made to scale.

Example



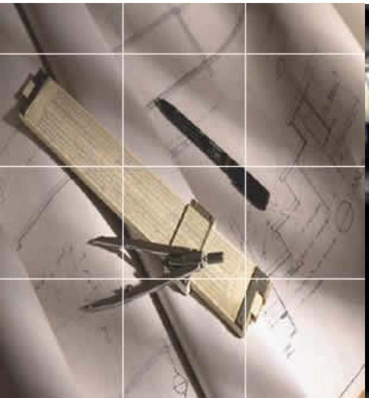
Computer drawing

The drawings are usually made by commercial software such as AutoCAD, solid works etc.

Example

Engineering Drawing

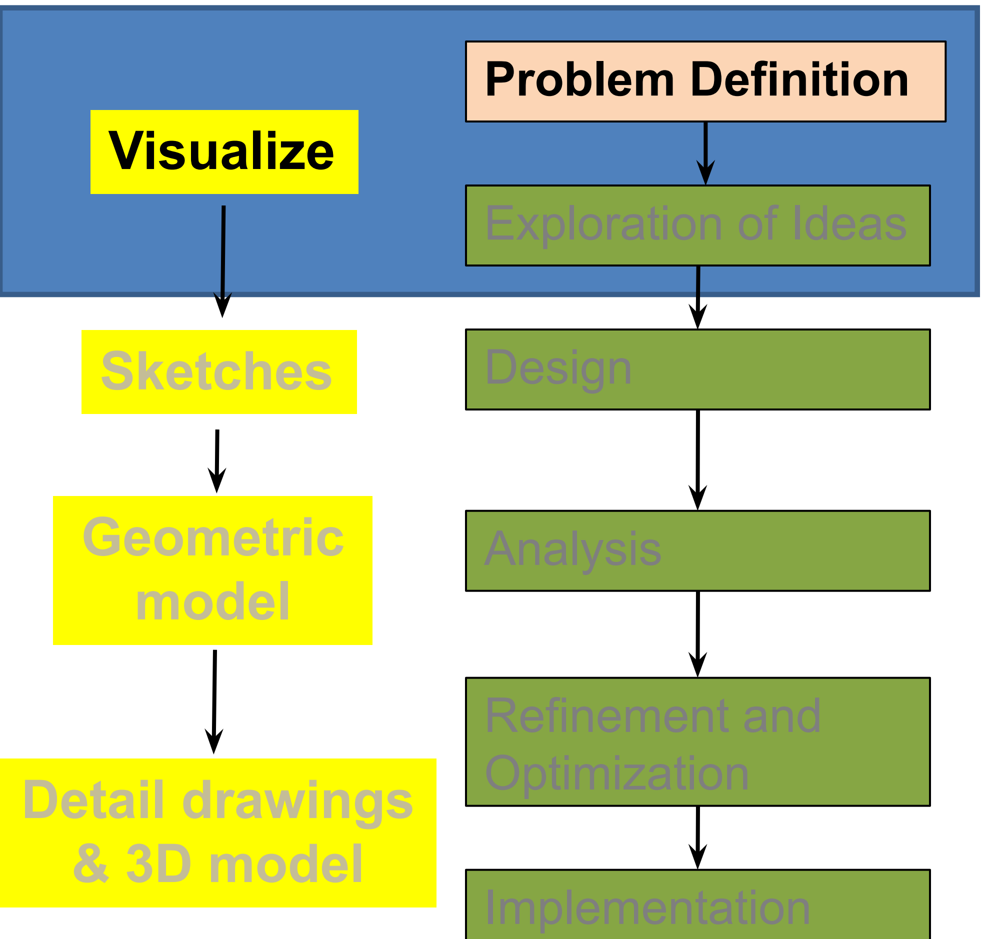
Process

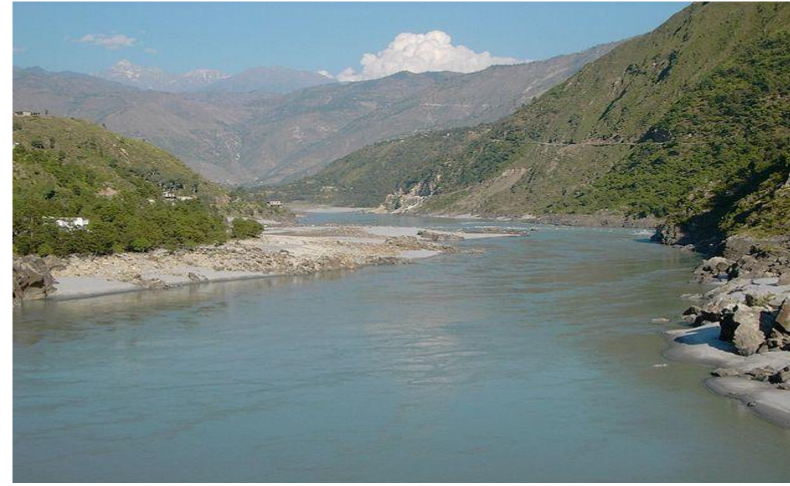


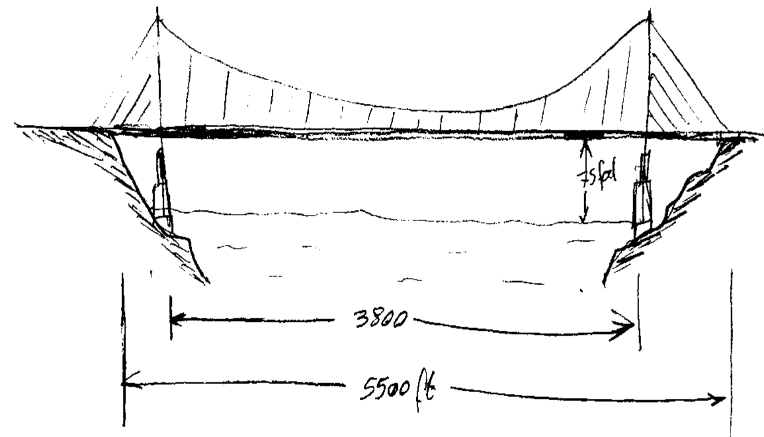

Engineering Drawing
In Design Process

Visualize	Problem identification & exploration of ideas	**Visualization** is the ability to mentally picture things that do not exist
Sketches	Draw , design and record initial ideas	**Communication** the design solution should be communicated to others without ambiguity
Geometric model	Created from sketches used or analysis	
Detail drawings & 3D model	Refine, optimize & record the precise data for production process or implementation	**Documentation** permanent record of the solution

Engineering Drawing
In Design Process







Engineering Drawing
In Design Process

Visualize

Problem Definition

Exploration of Ideas

Sketches

Design

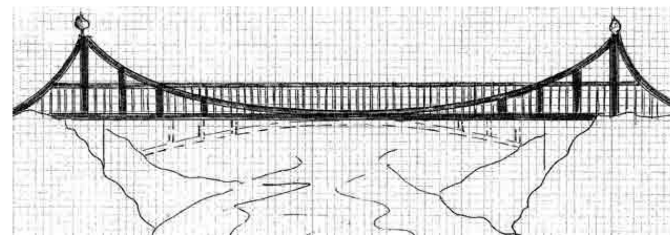
Geometric model

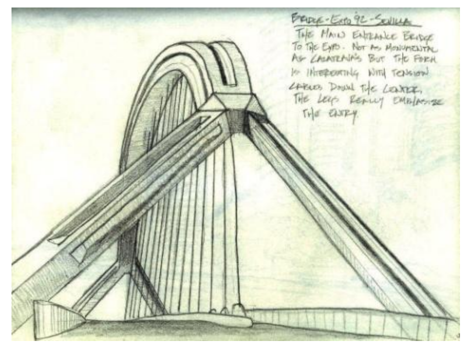
Analysis

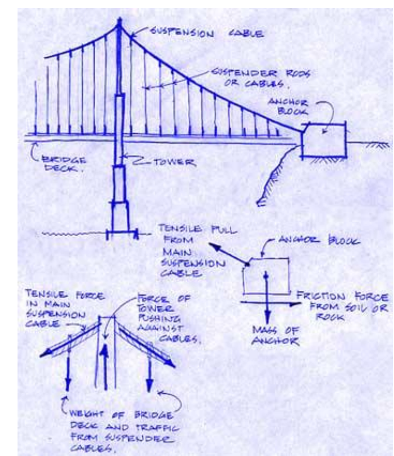
Refinement and Optimization

Detail drawings & 3D model

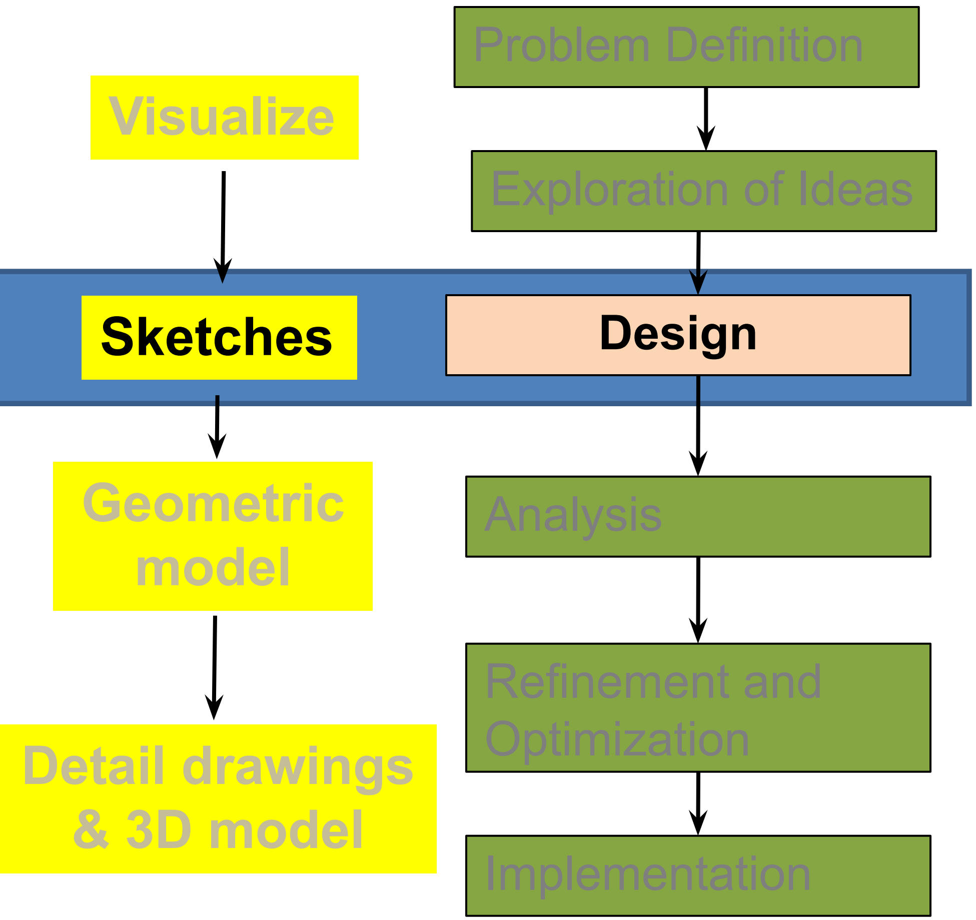
Implementation

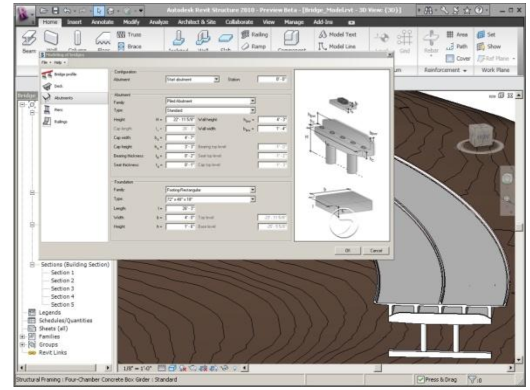


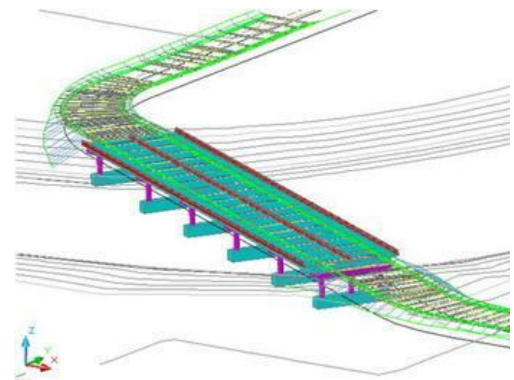


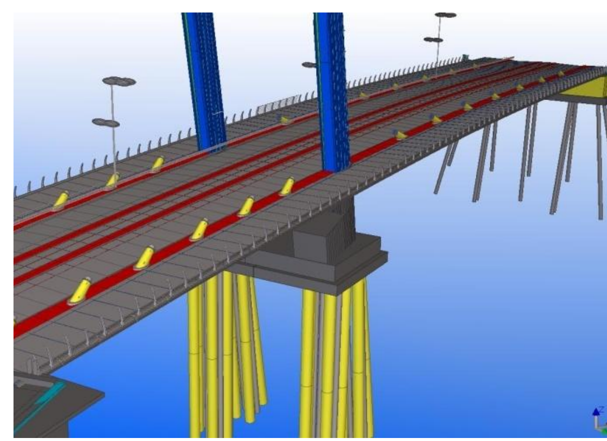


Engineering Drawing
In Design Process









Engineering Drawing
In Design Process

Visualize → Sketches → **Geometric model** → Detail drawings & 3D model

Problem Definition → Exploration of Ideas → Design → **Analysis** → Refinement and Optimization → Implementation

Engineering Drawing
In Design Process

Visualize

Sketches

Geometric model

Detail drawings & 3D model

Problem Definition

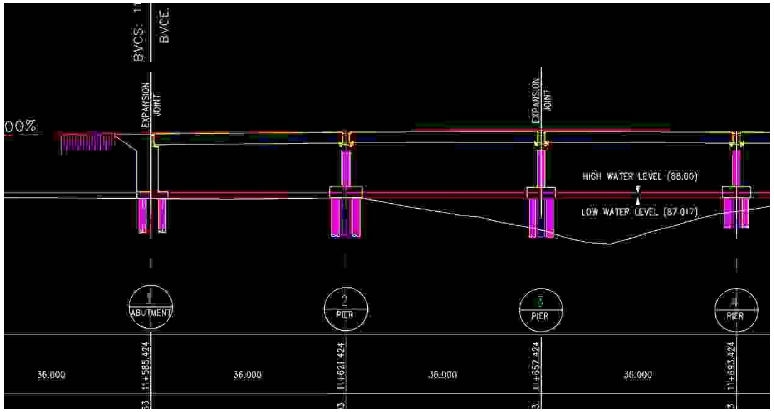
Exploration of Ideas

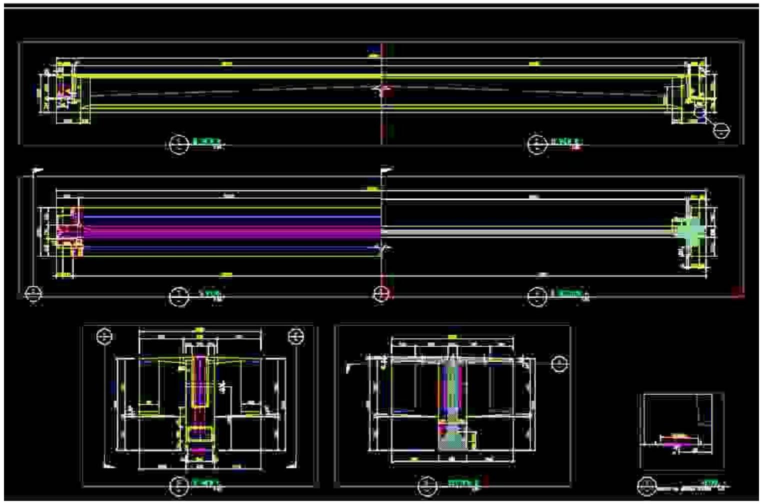
Design

Analysis

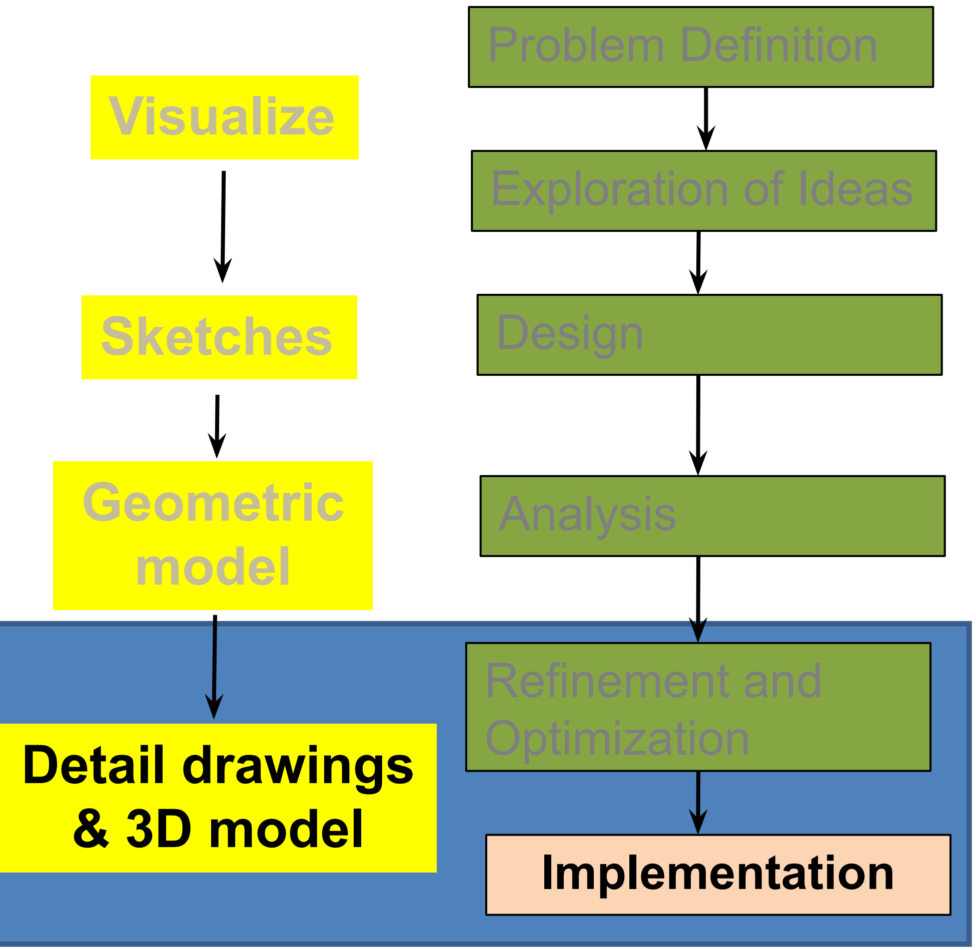
Refinement and Optimization

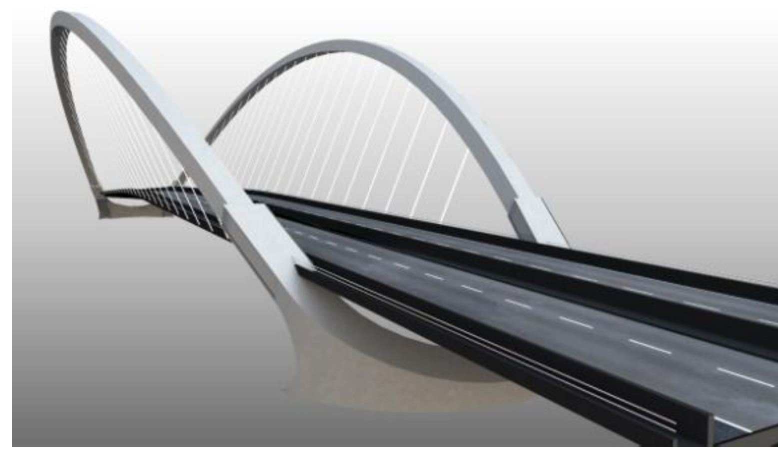
Implementation





Engineering Drawing
In Design Process





Engineering Drawing

Elements of Engineering Drawing

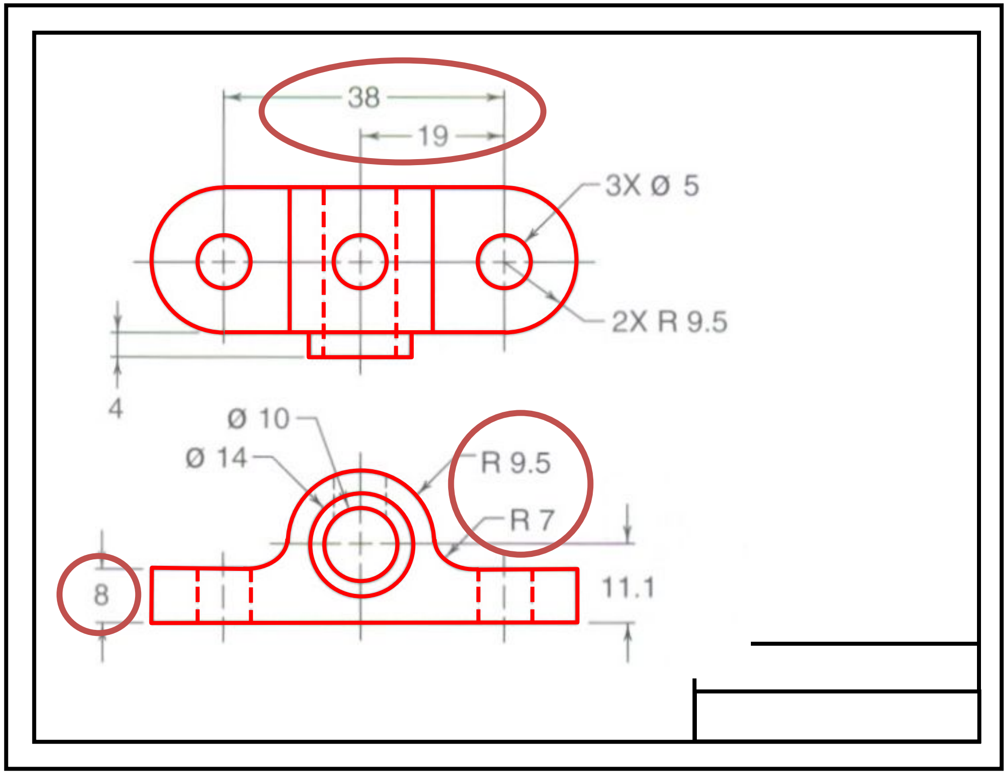
Engineering drawings are made up of ***graphics language*** and ***word language***.

Graphics language

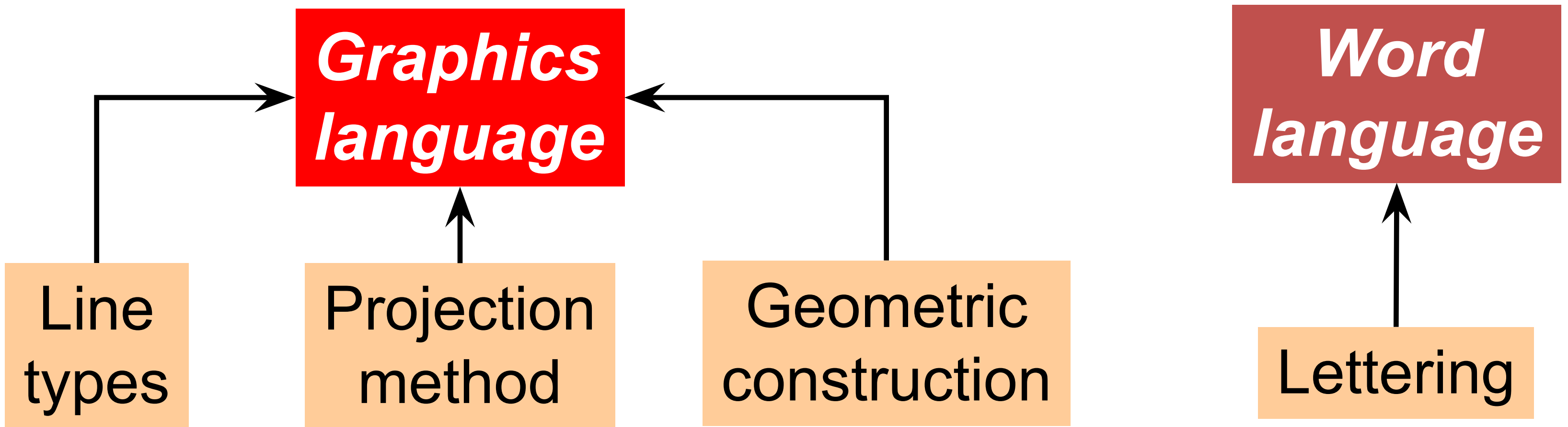
Describe a shape (mainly).

Word language

Describe size, location and specification of the object.

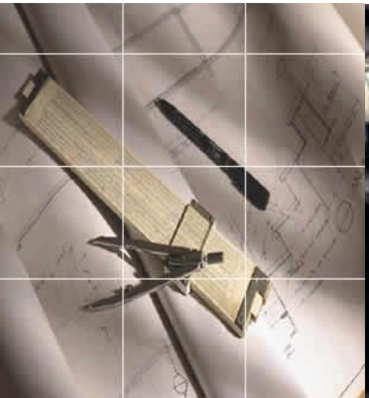


Basic Knowledge for Drafting



Engineering Drawing

Standards




Standards Definition

Drawing standards are *set of rules* that govern how technical drawings are represented.

Why Necessary

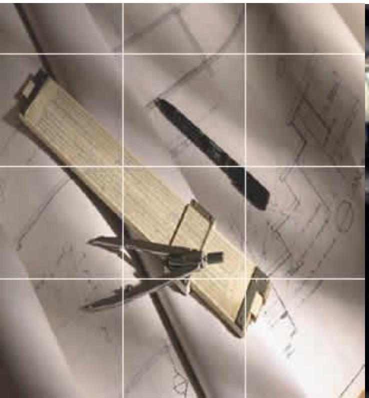
Drawing standards are used so that drawings *convey the same meaning* to everyone who reads them.

Standard Code

Country	Code	Full name
USA	ANSI	American National Standard Institute
Japan	JIS	Japanese Industrial Standard
UK	BS	British Standard
Australia	AS	Australian Standard
Germany	DIN	Deutsches Institut für Normung
	ISO	International Standards Organization

Engineering Drawing

Drawing Sheet



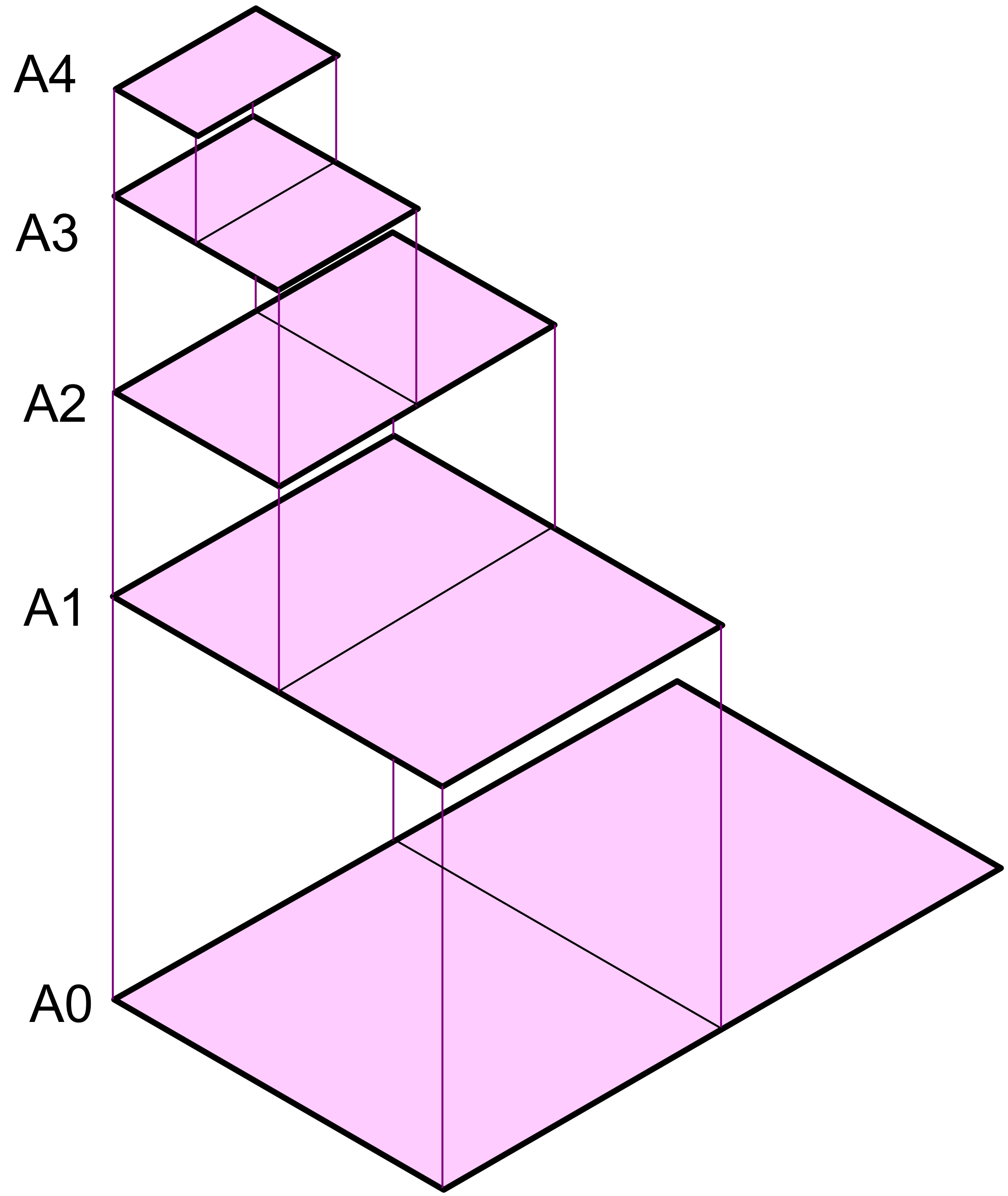

Drawing Sheet : Standard size

- Trimmed paper of a size A0 ~ A4.

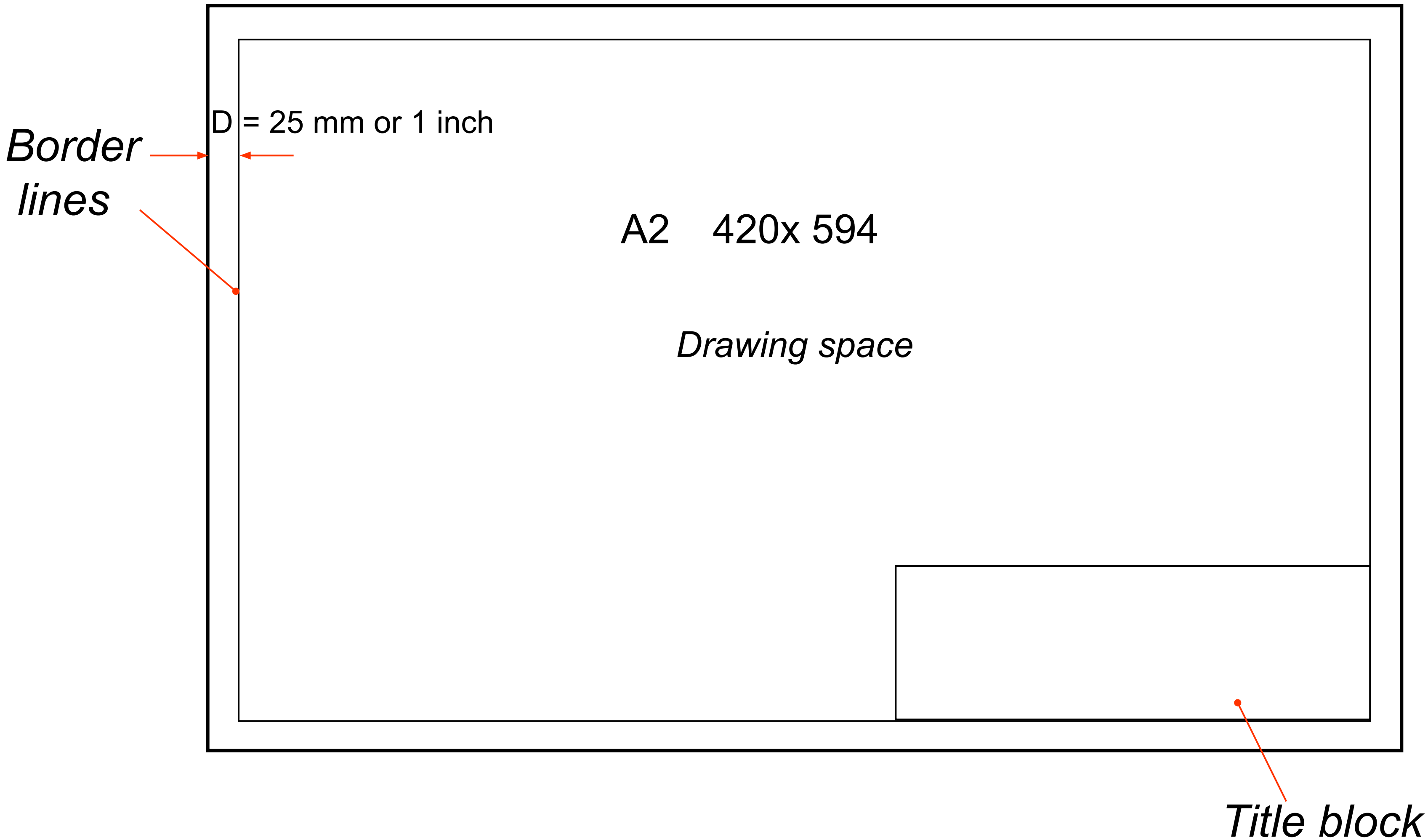
- Standard sheet size

A4	210 x 297
A3	297 x 420
A2	420 x 594
A1	594 x 841
A0	841 x 1189

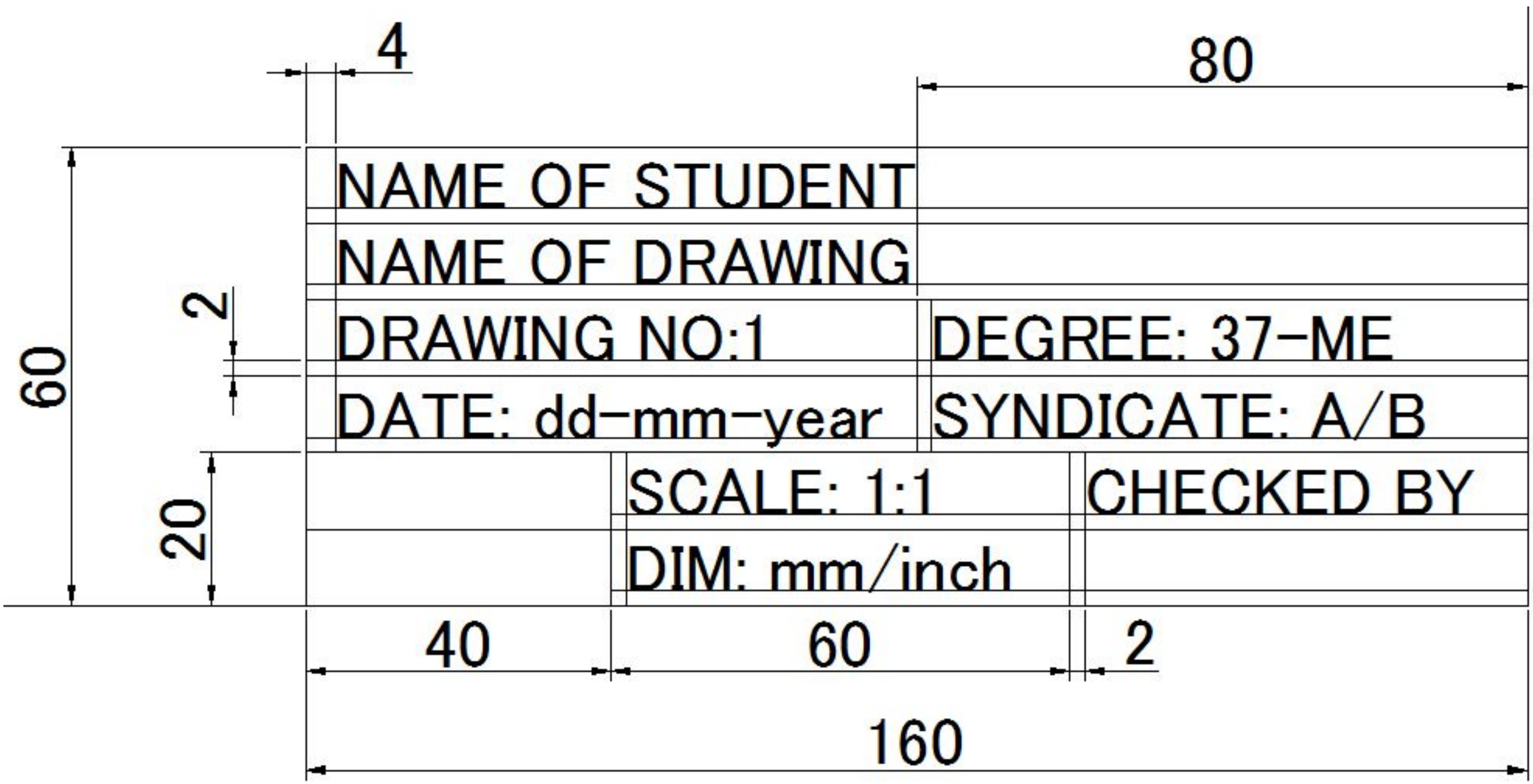
(Dimensions in millimeters)



Drawing Sheet : Orientation & Margin



Title Block (This course)

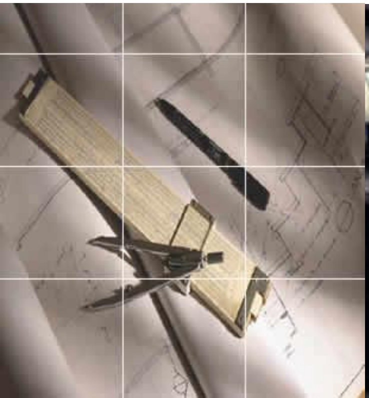


Title Block- Finished (This course)

NAME OF STUDENT			
NAME OF DRAWING			
DRAWING NO:1		DEGREE: 37-ME	
DATE: dd-mm-year		SYNDICATE: A/B	
	SCALE: 1:1		CHECKED BY
	DIM: mm/inch		

Engineering Drawing

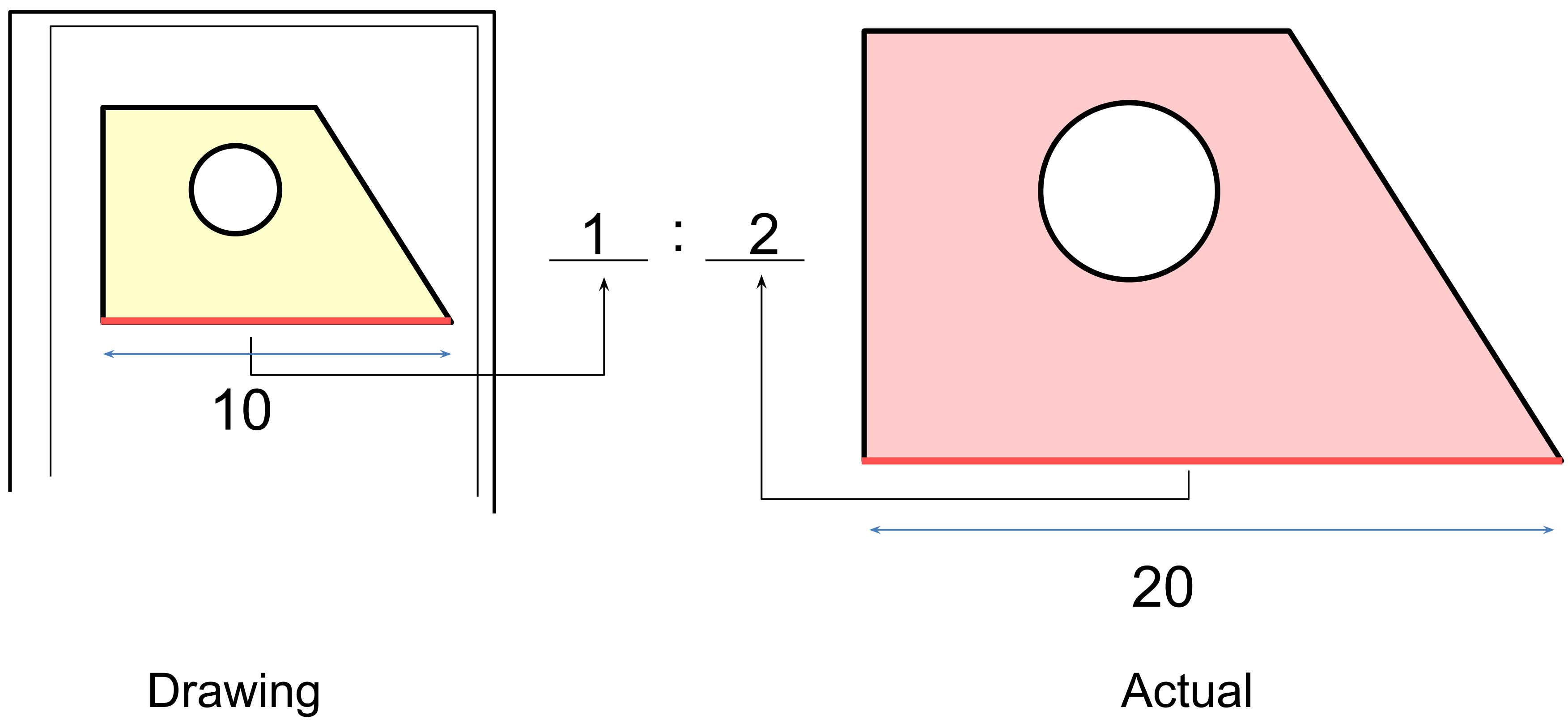
Drawing Scale

Drawing Scales : Definition

Length, size

Scale is a ratio between the linear dimension of a drawn representation of an object and the actual object.



Drawing Scales : Designation

Designation of a scale consists of the word "SCALE" followed by the indication of its ratio, as follows

SCALE 1:1 for full size

SCALE **X**:1 (X > 1) for an ***enlargement*** scales

SCALE 1:**X** (X > 1) for a ***reduction*** scales

Drawing scale is commonly found in a title block.

Drawing Scales : Standard scale

Standard reducing scales are

1:2, 1:5, 1:10, 1:20, 1:50, 1:100

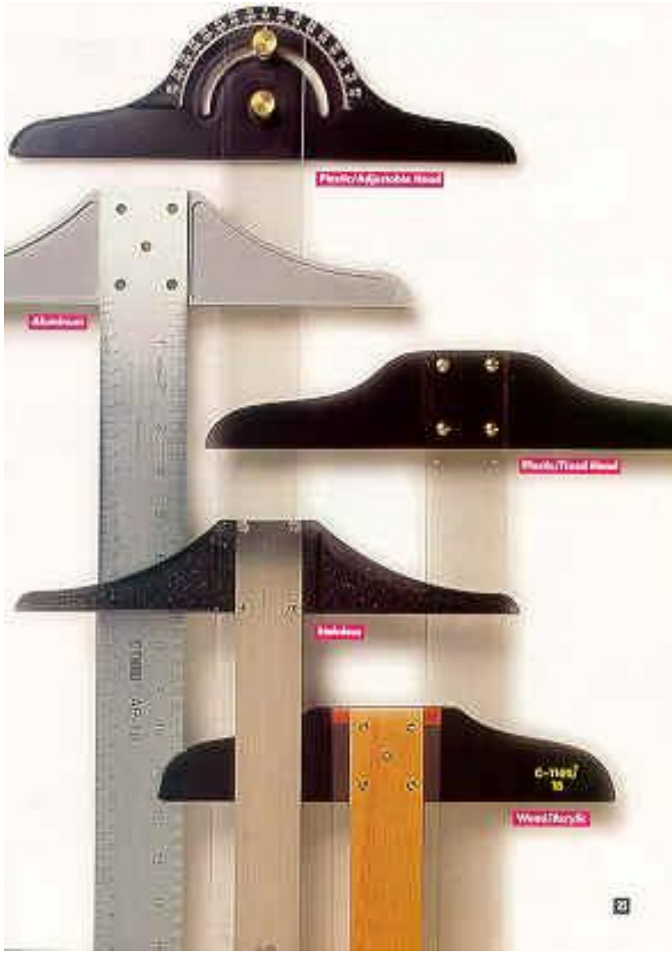
Standard enlarging scales are

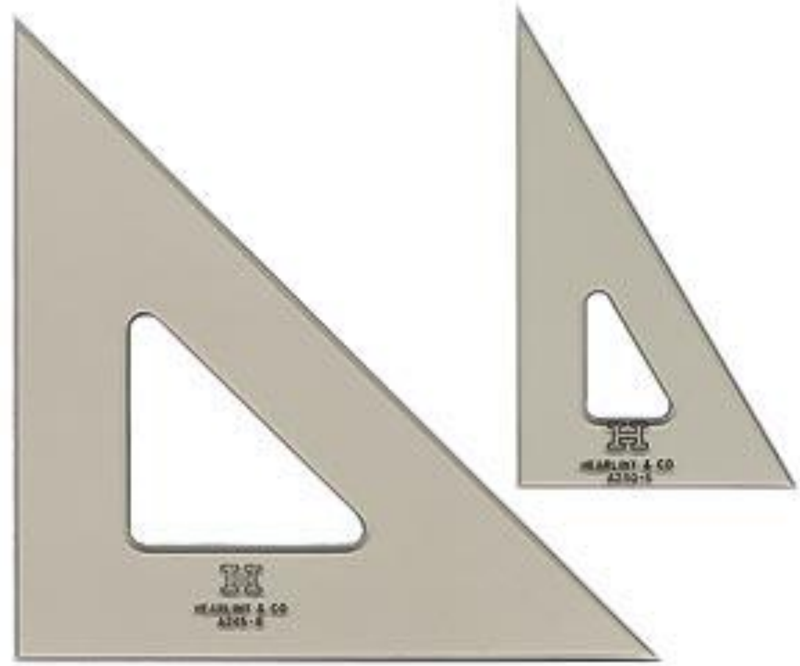
2:1, 5:1, 10:1, 20:1, 50:1, 100:1

Traditional Drawing Tools

DRAWING TOOLS

DRAWING TOOLS





1. T-Square

2. Triangles

DRAWING TOOLS

2H or HB for thick line

4H for thin line

3. Adhesive Tape

4. Pencils

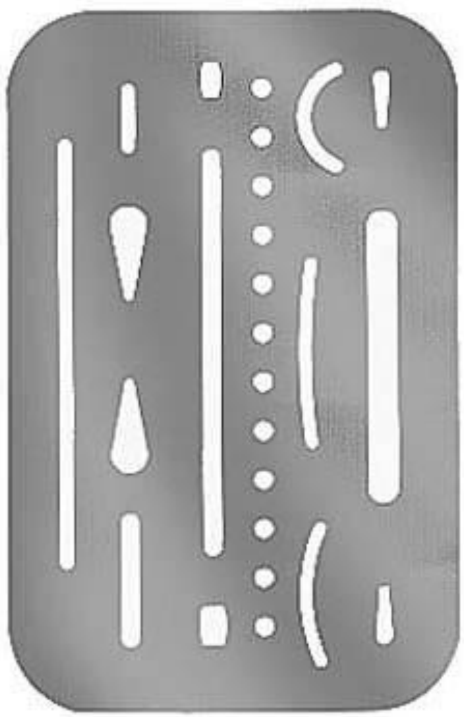
DRAWING TOOLS

5. Sandpaper

6. Compass

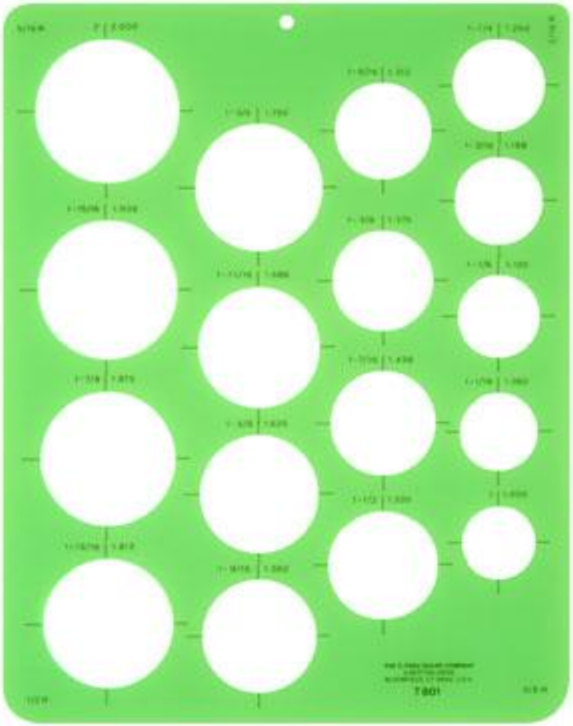
DRAWING TOOLS

7. Pencil Eraser



8. Erasing Shield

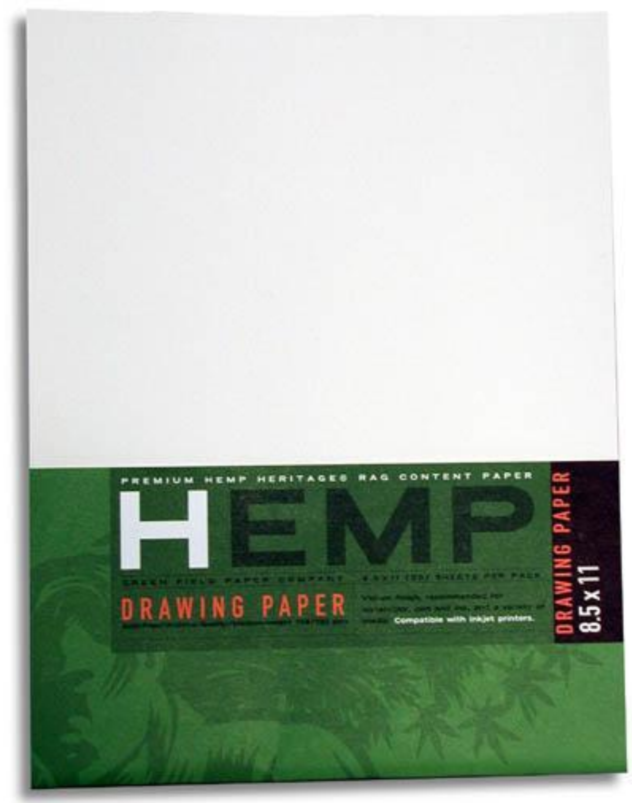
DRAWING TOOLS



9. Circle Template

10. Tissue paper

DRAWING TOOLS



11. Sharpener

12. Clean paper